

Quantum Computing
Post-Quantum Cryptography
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A quantum computer uses **quantum bits** and relies on of **quantum-mechanical phenomena** to perform computation

1. Brute-forcing n-bits key with Grover's algorithm would take **$2^{n/2}$**

➔ **Using symmetric encryption is still safe**

2. Factoring prime numbers with Shor's algorithm would be done in polynomial time

➔ **Using asymmetric encryption (key exchange and digital signatures) is at risk**

Post-Quantum Cryptography

Cryptographic schemes that can defeat quantum computers

- ➡ Still in research (started around 2006)
- ➡ On August 2024, the NIST released final versions of the first three Post Quantum Crypto Standards
- ➡ On November 2024, the NIST has announced prohibiting classical cryptography (RSA, DSA, ECDSA, ECDH) after 2035

	ML-KEM (Module-Lattice Key Encapsulation)	ML-DSA (Module-Lattice Digital Signature)	SLH-DSA (Stateless Hash-Based Signature)
Standard	FIPS-203	FIPS-204	FIPS-205
Purpose	Key Exchange to generate a 256-bits shared key	Digital Signature Scheme that offers a good balance of performance relative to signature sizes	Digital Signature Scheme designed as a fallback / conservative option with smaller public keys but large signatures and slower performances
Description	Exists in three versions ML-KEM-512 ML-KEM-768 ML-KEM-1024	Exists in three versions ML-DSA-44 ML-DSA-65 ML-DSA-87	Exists in 12 variants as combinations of different key sizes (128, 192, 256), different hash (Sha2, Shake) and different algorithm trade-off (small vs fast)
Algorithm and Mathematical Foundation	Module Learning With Errors (MLWE) based on a lattice approach	CRYSTALS-Dilithium based on lattice signature	SPHINCS+ a purely hash-based signature scheme, does not rely on a lattice approach

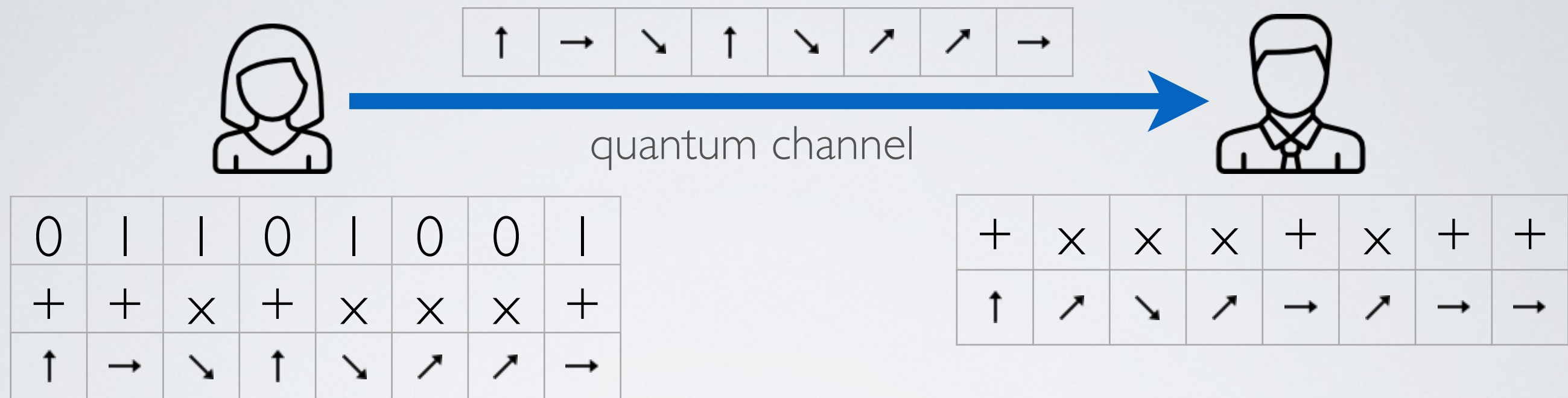
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The use uses quantum bits and quantum-mechanical phenomena to realize cryptographic tasks

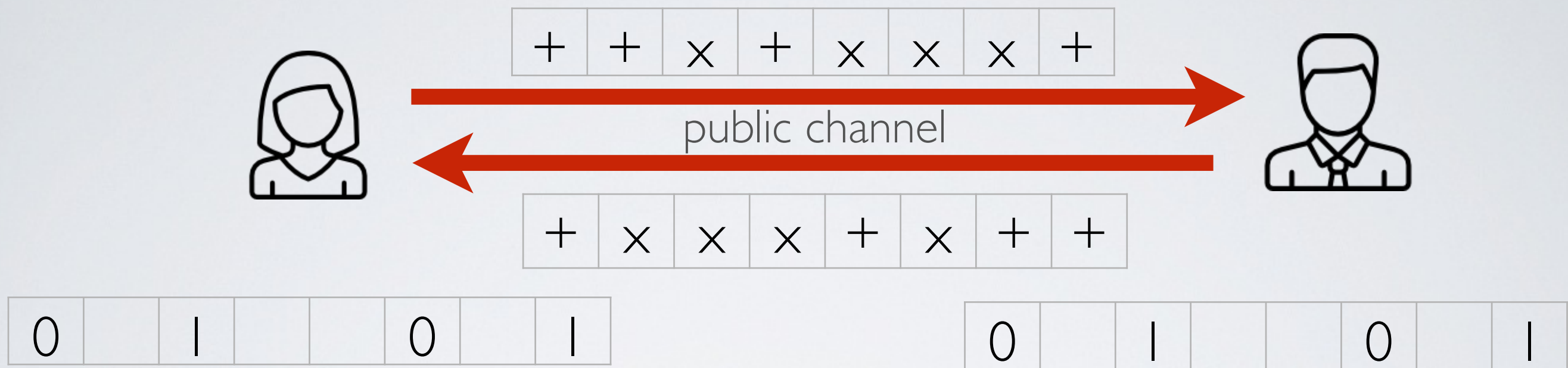
- ➡ Example : Quantum Key Distribution - use a quantum channel to establish a shared key to use on a public channel

Quantum Key Distribution - step I



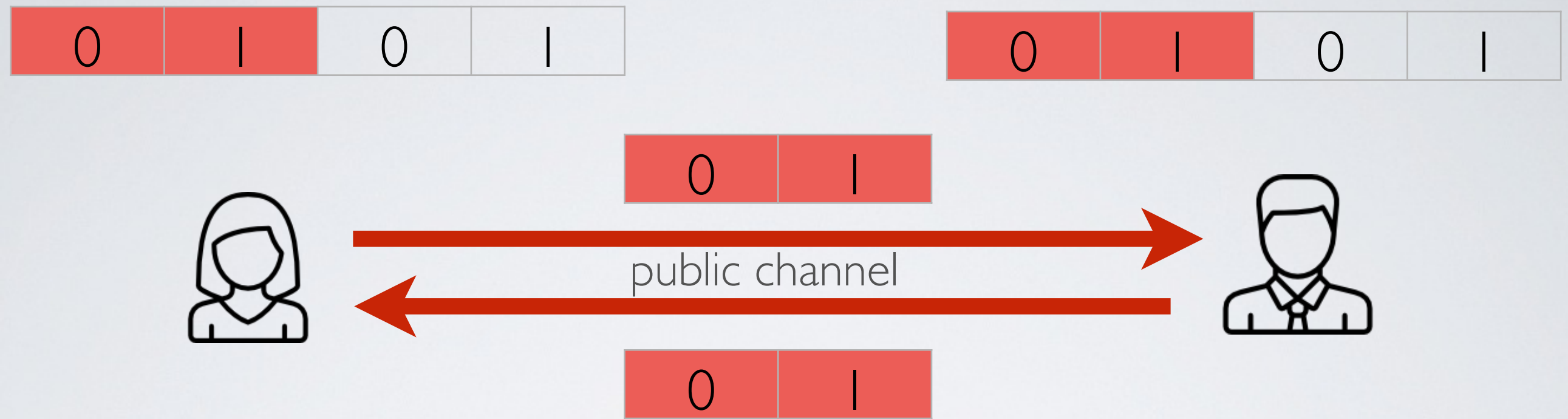
- I. Alice creates:
 - I. a sequence of random sequence of bits
 - II. a sequence of random sequence of basis
 - III. a sequence of random sequence of polarized photons corresponding to the basis
2. Alice sends the photon sequence to Bob over the quantum channel
3. Bob selects a random sequence of basis
4. Bob measures Alice's sequence of photons using his basis

Quantum Key Distribution - step 2



5. Alice and Bob exchange their sequence of basis on the public channel
6. The basis that are commonly correct are used to generate the key

Quantum Key Distribution - step 3



Has Eve eavesdrop on the quantum Channel ?

- ➔ Eavesdropping the quantum channel modifies the polarization of the photons
- 7. Alice and Bob spare and exchange a sub sequence of their shared secret key
- 8. If this subsequence match, it means that nobody has eavesdrop the quantum channel. If not, the key is invalid.